Low-degree superpolynomial and high-degree exponential lower bounds for sums of ordered set-multilinear ABPs

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Allowed standard prerequisites (editorial).

- Relative-rank properties and inner-product gadget. Author Claim clm:rk-props and Observation obs:ip-full-rank; exact statements retained.
- Explicit VP hard polynomial with full relative rank. Author Lemma lem:Fn-VP-props explicitly cites KushS23; source construction and full quoted guarantee retained.
- Standard multiplicative Chernoff bound. Author Theorem thm:Chernoff explicitly cites Mitzenmacher-Upfal; exact source statement retained.

Source reference. Original section 1.3: number-theory/sources/ccc-2024-20/native/main.tex, source lines 150–150. Optional introductory pointer to results; counted full target statements and proofs are included.

An *algebraic circuit* over a field $\mathbb{F}$ for a multivariate polynomial $P(x_1, \ldots, x_N)$ is a directed acyclic graph (DAG) whose internal vertices (called gates) are labeled as either + (sum) or × (product), and leaves (vertices of in-degree zero) are labeled by the variables x_i or constants from $\mathbb{F}$. A special output gate (the root of the DAG) represents the polynomial P. If the DAG happens to be a tree, such a resulting circuit is called an *algebraic formula*. The size of a circuit or formula is the number of nodes in the DAG. We also consider the product-depth of the circuit, which is the maximum number of product gates on a root-to-leaf path. The class VP (respectively, VF) is then defined to be the collection of all polynomials having at most polynomially large degree which can be computed by polynomial-sized circuits (respectively, formulas).

An *algebraic branching program* (ABP) is a layered DAG with two special nodes in it: a start-node and an end-node. All edges of the ABP go from layer $\ell - 1$ to layer ℓ for some ℓ (say start-node is the unique node in layer 0 and end-node is the unique node in the last layer) and are labeled by a linear polynomial. Every directed path γ from start-node to end-node computes the monomial P_γ, which is the product of all labels on the path γ. The ABP computes the polynomial $P = \sum_\gamma P_\gamma$, where the sum is over all paths γ from start-node to end-node. Its size is simply the number of nodes in the DAG, its *depth* is the length of the longest path from the start-node to the end-node, and *width* is the maximum number of nodes in any layer. The class VBP is then defined to be the collection of all polynomials (with polynomially-bounded degree) which can be computed by polynomial-sized branching programs. ABPs are known to be of intermediate complexity between formulas and circuits; in other words, we know the inclusions $\mathsf{VF} \subseteq \mathsf{VBP} \subseteq \mathsf{VP} \subseteq \mathsf{VNP}$.

A polynomial is said to be *homogeneous* if each monomial has the same total degree and *multilinear* if every variable occurs at most once in any monomial. Now, suppose that the underlying variable set is partitioned into d sets $X_1, \ldots, X_d$. Then the polynomial is said to be *set-multilinear* with respect to this variable partition if each monomial in P has *exactly* one variable from each set. Note that a set-multilinear polynomial is both multilinear and homogeneous, and has degree precisely d if it is set-multilinear over d sets. Next, we define different models of computation corresponding to these variants of polynomials classes. An algebraic formula/branching program/circuit is set-multilinear with respect to a variable partition $(X_1, \ldots, X_d)$ if each internal node in the formula/branching program/circuit computes a set-multilinear polynomial.[1] Multilinear and homogeneous formulas/branching programs/circuits are defined analogously.

Definition 0.1 (Ordered smABP)**.** Given a variable partition $(X_1, \ldots, X_d)$, we say that a set-multilinear branching program of depth d is said to be *ordered* with respect to an ordering (or permutation) $\sigma \in S_d$ if for each $\ell \in [d]$, all edges of the ABP from layer $\ell - 1$ to layer ℓ are labeled using a linear form over the variables in $X_{\sigma(\ell)}$. It is simply said to be *ordered* if there *exists* an ordering σ such that it is ordered with respect to σ.

At this point, it is essential to take note of the terminology in this context: in this paper, a general (or "unordered") set-multilinear branching program refers to an ABP for which each internal node computes a polynomial that is set-multilinear with respect to *some* subset of the global partition, whereas an *ordered* set-multilinear branching program is more specialized and has the property that any two nodes in the same layer compute polynomials that are set-multilinear with respect to the *same* partition.

Definition 0.2 ($\sum$smABP)**.** Given a polynomial $P(X)$ that is set-multilinear with respect to the variable partition $X = (X_1, \ldots, X_d)$, we say that $\sum_{i=1}^{t} A_i$ is a $\sum$smABP computing P if indeed $\sum_{i=1}^{t} A_i(X) = P(X)$, and each A_i is an ordered set-multilinear branching program i.e., each A_i is ordered with respect to some $\sigma_i \in S_d$. We call t (i.e., the number of summands in a $\sum$smABP) its

[1] Of course, a non-root node need not be set-multilinear with respect to the *entire* variable partition. Nevertheless, here we demand that it must be set-multilinear with respect to some *subset* of the collection $\{X_1, \ldots, X_d\}$.

support size and define its *max-width* and *total-width* to be the maximum over the width of each A_i and the sum of the width of each A_i, respectively.

Theorem 0.3 ("Low"-Degree $\sum$smABP Lower Bounds)**.** *Let $d \leq n$ be growing parameters satisfying $d = \omega(\log n)$. There is a $\Theta(dn)$-variate degree d set-multilinear polynomial $F_{n,d}$ in VP such that $F_{n,d}$ cannot be computed by a $\sum$smABP of total-width $\mathrm{poly}(n)$.*

Theorem 0.4 (Exponential Lower Bounds for $\sum$smABP)**.** *There is a set-multilinear polynomial $F_{n,n}$ in VP, in $\Theta(n^2)$ variables and of degree $\Theta(n)$, such that any $\sum$smABP computing $F_{n,n}$ requires total-width $\exp(\Omega(n^{1/3}))$.*

1 Preliminaries

1.1 Relative Rank and its Properties

We first describe the notation that we need to define the measures that we use to prove our results described in Section 1.3. Instead of directly working with the rank of the partial derivative matrix, we work with the following normalized form.

Definition 1.1. Let $w = (w_1, w_2, \ldots, w_d)$ be a tuple (or word) of non-zero real numbers. For a subset $S \subseteq [t]$, we shall refer to the sum $\sum_{i \in S} w_i$ by w_S, and by $w|_S$, we will refer to the tuple obtained by considering only the elements of w that are indexed by S. Given a word $w = (w_1, \ldots, w_d)$, we denote by $\overline{X}(w)$ a tuple of d sets of variables $(X(w_1), \ldots, X(w_d))$ where $|X(w_i)| = 2^{|w_i|}$.[2] We denote by $\mathbb{F}_{\mathrm{sm}}[\mathcal{T}]$ the set of set-multilinear polynomials over the tuple of sets of variables $\mathcal{T}$.

Definition 1.2 (Relative Rank Measure of [LST21])**.** Let $\overline{X} = (X_1, \ldots, X_d)$ be a tuple of sets of variables such that $|X_i| = n_i$ and let $f \in \mathbb{F}_{\mathrm{sm}}[\overline{X}]$. Let $w = (w_1, w_2, \ldots, w_d)$ be a tuple (or word) of non-zero real numbers such that $2^{|w_i|} = n_i$ for all $i \in [d]$. Corresponding to a word w, define $\mathcal{P}_w \coloneqq \{i \mid w_i > 0\}$ and $\mathcal{N}_w \coloneqq \{i \mid w_i < 0\}$. Let $\mathcal{M}_w^{\mathcal{P}}$ be the set of all set-multilinear monomials over the subset of the variable sets $X_1, X_2, \ldots, X_d$ precisely indexed by $\mathcal{P}_w$, and similarly let $\mathcal{M}_w^{\mathcal{N}}$ be the set of all set-multilinear monomials over these variable sets indexed by $\mathcal{N}_w$.

Define the 'partial derivative matrix' matrix $\mathcal{M}_w(f)$ whose rows are indexed by the elements of $\mathcal{M}_w^{\mathcal{P}}$ and columns indexed by the elements of $\mathcal{M}_w^{\mathcal{N}}$ as follows: the entry of this matrix corresponding to a row m_1 and a column m_2 is the coefficient of the monomial $m_1 \cdot m_2$ in f. We define

$$\mathrm{relrk}_w(f) \coloneqq \frac{\mathrm{rank}(\mathcal{M}_w(f))}{\sqrt{|\mathcal{M}_w^{\mathcal{P}}| \cdot |\mathcal{M}_w^{\mathcal{N}}|}} = \frac{\mathrm{rank}(\mathcal{M}_w(f))}{2^{\frac{1}{2}\sum_{i \in [d]} |w_i|}}.$$

The following is a simple result that establishes various useful properties of the relative rank measure.

Claim 1.3 ([LST21])**.** 1. (Imbalance) Say $f \in \mathbb{F}_{\mathrm{sm}}[\overline{X}(w)]$. Then, $\mathrm{relrk}_w(f) \leq 2^{-|w_{[d]}|/2}$.

[2] In particular, $2^{|w_i|} \in \mathbb{N}$.

2. (Sub-additivity) If $f, g \in \mathbb{F}_{\mathrm{sm}}[\overline{X}(w)]$, then $\mathrm{relrk}_w(f+g) \leq \mathrm{relrk}_w(f) + \mathrm{relrk}_w(g)$.

3. (Multiplicativity) Say $f = f_1 f_2 \cdots f_t$ and assume that for each $i \in [t]$, $f_i \in \mathbb{F}_{\mathrm{sm}}[\overline{X}(w|_{S_i})]$, where $(S_1, \ldots, S_t)$ is a partition of $[d]$. Then

$$\mathrm{relrk}_w(f) = \prod_{i \in [t]} \mathrm{relrk}_{w|_{S_i}}(f_i).$$

We will repeatedly make use of the following.

Theorem 1.4 (Chernoff Bound, as stated in [MU05])**.** *Suppose $X_1, \ldots, X_n$ are independent random variables taking values in $\{0,1\}$. Let X denote their sum and let $\mu = \mathbb{E}[X]$ denote the expected value of the sum. Then for any $\delta > 0$,*

$$\begin{aligned}
\mathbb{P}[X \geq (1+\delta)\mu] &\leq \exp\left(-\frac{\delta^2 \mu}{2+\delta}\right), \text{ if } 0 \leq \delta, \\
\mathbb{P}[X \leq (1-\delta)\mu] &\leq \exp\left(-\frac{\delta^2 \mu}{2}\right), \text{ if } 0 < \delta < 1, \\
\mathbb{P}[|X-\mu| \geq \delta\mu] &\leq 2 \cdot \exp\left(-\frac{\delta^2 \mu}{3}\right), \text{ if } 0 < \delta < 1.
\end{aligned}$$

1.2 Inner Product Gadget

The following observation is used crucially to construct the hard polynomials in VP as well as VBP.

Observation 1.5 ([KS23])**.** Let $n = 2^k$ and $X_1 = \{x_{1,1}, \ldots, x_{1,n}\}$ and $X_2 = \{x_{2,1}, \ldots, x_{2,n}\}$ be two disjoint sets of variables. Then, for any *symmetric* word $w \in \{k, -k\}^2$ (i.e., where $w_1 + w_2 = 0$) and for the inner product 'gadget' $f = X_1 \cdot X_2 = \sum_{i=1}^{n} x_{1,i} x_{2,i}$, $\mathrm{relrk}_w(f) = 1$ i.e., $\mathcal{M}_w(f)$ is full-rank.

1.3 A Hard Set-multilinear Polynomial in VP

Let d be an even integer and let $X = (X_1, \ldots, X_d)$ be a collection of sets of variables where each $|X_i| = n$, and similarly, let $Y = (Y_1, \ldots, Y_d)$ be a distinct collection of sets of variables where each $|Y_i| = n$. We shall refer to the Y-variables as the *auxiliary* variables. For i and $j \in \{1, \ldots, d\}$, let $X_i \cdot X_j$ denote the inner-product quadratic form $\sum_{k=1}^{n} x_{ik} x_{jk}$. Here, we shall assume that $X_i = \{x_{i,1}, \ldots, x_{i,n}\}$ and $Y_i = \{y_{i,1}, \ldots, y_{i,n}\}$.

For two integers $i \in \mathbb{N}$ and $j \in \mathbb{N}$, we denote $[i,j] = \{k \in \mathbb{N} : i \leq k \text{ and } k \leq j\}$ and call such a set an *interval*. For every interval $[i,j] \subseteq [d]$, we define a polynomial $f_{i,j}(X, Y) \in \mathbb{F}_{\mathrm{sm}}[X_i, \ldots, X_j, Y_i, \ldots, Y_j]$ as follows:

$$f_{i,j} = \begin{cases} y_{i,j} y_{j,i} (X_i \cdot X_j) & \text{if } j = i+1 \\ 0 & \text{if } j - i \text{ is even} \\ y_{i,j} y_{j,i} (X_i \cdot X_j) \cdot f_{i+1,j-1} + \sum_{r=i+1}^{j-1} f_{i,r} f_{r+1,j} & \text{otherwise} \end{cases}$$

These $f_{i,j}$ in present form were defined in [KS23], but were in turn inspired from an earlier work of Raz and Yehudayof ([RY08]) in the multilinear context. [KS23] shows that they have the following full-rank property that will be instrumental for us.

Lemma 1.6 ([KS23])**.** *Let $n = 2^k$ and $d \leq n$ be an even integer. Over any field $\mathbb{F}$ of characteristic zero, the polynomial $F_{n,d} = f_{1,d} \in \mathbb{F}_{\mathrm{sm}}[X, Y]$ as defined above satisfies the following: For any $w \in \{-k, k\}^d$ with $w_{[d]} = 0$, $\mathcal{M}_w(F_{n,d})$ is full-rank when viewed as a matrix over the field $\mathbb{F}(Y)$, the field of rational functions over the Y variables.*

2 Separation between VP and $\sum$smABP

In the theorem below, $F_{n,d}$ refers to the polynomial defined in Section 1.3. In this section, we prove Theorem 0.4 and Theorem 0.3 by first proving the following statement.

Lemma 2.1. *Given large enough integers $d \leq n$, any $\sum$smABP of max-width s and support size t computing $F_{n,d}(X_1, \ldots, X_d, Y_1 \ldots, Y_d)$ must satisfy at least one of the following:*

- *either $t > e^{d/96}$,*
- *or $t \cdot s^q \geq n^{\frac{\sqrt{dq}}{20}}$, for any integer q in the range $[12 \ln(2t\sqrt{d}), d/4]$.*

Proof. Suppose that $t \leq e^{d/96}$ (so that the range in the theorem statement is indeed well-defined).

First, we observe that for any $\sum$smABP computing $F_{n,d}(X, Y)$, we can view each summand as an ordered set-multilinear branching program with respect to only the X variables. In other words, by appropriately collapsing the layers labelled using the Y variables, each summand is a set-multilinear branching program over the field $\mathbb{F}(Y)$ *ordered* with respect to $(X_{\sigma(1)}, \ldots, X_{\sigma(d)})$ for some permutation $\sigma \in S_d$. It is easy to see that this collapsing process does not increase the width or the size of any summand in the branching program. The edge labels do get altered however: the coefficients of the X variables in any edge label can now be polynomials in the Y variables (and therefore, field constants in $\mathbb{F}(Y)$).

Let A be such a set-multilinear branching program of width s and depth d that is *ordered* with respect to $(X_1, \ldots, X_d)$[3]. Recall that this means that for each $\ell \in [d]$, all edges of the ABP from layer $\ell - 1$ to layer ℓ are labeled using a linear form in X_ℓ. Given a node u in layer i and a node v in layer $j > i$ of A, define $g_{u,v}$ to be the polynomial computed by the ABP restricted to the layers $i + 1, \ldots, j - 1$ with the source and the sink defined by u and v respectively. Consider an integer q in the range specified in the theorem statement and let r be the largest integer r such that the product $qr < d$ i.e., we must have $d - qr \leq q$. Consider the following decomposition of A:

$$A(X_1, \ldots, X_d) = \sum_{u_1, \ldots, u_q} \prod_{i=1}^{q+1} g_{u_{i-1}, u_i},$$

[3]In general, A may be ordered with respect to an arbitrary permutation σ, but the assumption that σ is the identity permutation in the discussion that follows is without loss of generality.

where u_0 and u_{q+1} are defined to be the source and the sink of A respectively, and for $1 \le i \le q$, u_i varies over all choices of nodes in layer $r \cdot i$. Note that hence, this expression contains at most s^q terms. Also, note that each g_{u_{i-1},u_i} is set-multilinear over the partition $(X_i)_{i \in S_i}$ where S_i is the set $\{r(i-1)+1, \ldots, ri\}$[4] of length exactly r for $i \in [q]$, and $S_{q+1} = \{rq+1, \ldots, d\}$ has length at most q. We now analyze the relative rank of each summand.

Let $w \in \{-k, k\}^d$ (where $k = \log n$) be an arbitrary word. By Claim 1.3, we see that

$$\mathrm{relrk}_w\left(\prod_{i=1}^{q+1} g_{u_{i-1},u_i}\right) = \prod_{i=1}^{q+1} \mathrm{relrk}_{w|_{S_i}}(g_{u_{i-1},u_i}) \le \prod_{i=1}^{q} \mathrm{relrk}_{w|_{S_i}}(g_{u_{i-1},u_i}) \le \prod_{i=1}^{q} 2^{-|w_{S_i}|/2} = 2^{-\frac{1}{2}\left(\sum_{i=1}^{q} |w_{S_i}|\right)},$$

from which we observe that the task of upper bounding this rank can be reduced to the task of lower bounding the sum $\sum_{i=1}^{q} |w_{S_i}|$, which is established below.

Choose w from $\{-k, k\}^d$ (where $k = \log n$) uniformly at random. For each $i \in [q]$, let E_i denote the (bad) event that $|w_{S_i}| \le \sqrt{r}k/4$. Since S_i is an interval of length r, by a standard estimation of binomial coefficients, we obtain that $\mathbb{P}[E_i] \le 1/4$. Then, by the Chernoff bound (Theorem 1.4), the probability that at least half of the events E_i occur is at most $e^{-\frac{q}{12}}$. Therefore, with probability at least $1 - e^{-\frac{q}{12}}$,

$$\sum_{i=1}^{q} |w_{S_i}| \ge q\sqrt{r}k/8,$$

and therefore, by the sub-additivity of $\mathrm{relrk}_w(\cdot)$,

$$\mathrm{relrk}_w(A) \le s^q 2^{-\frac{kq\sqrt{r}}{16}} = s^q n^{-\frac{q\sqrt{r}}{16}}.$$

Now, let $\sum_{i=1}^{t} A_i$ be a $\sum$smABP computing $F_{n,d}$ with max-width bounded by s, and such that each A_i is ordered set-multilinear with respect to the variable partition $(X_{\sigma_i(1)}, \ldots, X_{\sigma_i(d)})$ for some permutation $\sigma_i \in S_d$. By the union bound and the discussion above, it follows that with probability at least $1 - t \cdot e^{-\frac{q}{12}}$,

$$\mathrm{relrk}_w(F_{n,d}) = \mathrm{relrk}_w\left(\sum_{i=1}^{t} A_i\right) \le t \cdot s^q n^{-\frac{q\sqrt{r}}{16}}.$$

But now, we can condition on the event that $w_{[d]} = 0$ (which occurs with probability $\Theta(\frac{1}{\sqrt{d}})$) to establish the existence of a word $w \in \{-k, k\}^d$ with $w_{[d]} = 0$ such that w satisfies $\mathrm{relrk}_w(P) \le t \cdot s^q n^{-q\sqrt{r}}$. This is because of the given bound $q \ge 12\ln(2t\sqrt{d})$. Because $\mathrm{relrk}_w(F_{n,d}) = 1$ for such a w by Lemma 1.6, we conclude that $t \cdot s^q \ge n^{\frac{q\sqrt{r}}{16}} \ge n^{\frac{\sqrt{(d-q)q}}{16}} \ge n^{\frac{\sqrt{dq}}{20}}$, where the last inequality follows from our choice of q.

□

Proof of Theorem 0.4. We invoke Lemma 2.1 with $d = n$. If $t \ge 2^{n^{1/3}}$, then we trivially have that the total-width is at least $\exp(\Omega(n^{1/3}))$, so assume $t \le 2^{n^{1/3}}$. We shall show that then, $s = \exp(\Omega(n^{1/3}))$, which will yield the desired result.

[4] If A is instead ordered with respect to σ, then S_i is taken to be the set $\{r(i-1)+1, \ldots, ri\}$.

Set $q = \lceil 15n^{1/3} \rceil$. Then clearly, $q \leq n/4$. Moreover, as $t \leq 2^{n^{1/3}}$ by assumption, we verify that

$$q = \lceil 15n^{1/3} \rceil > 12(n^{1/3} + \ln(2\sqrt{d})) \geq 12 \ln(2t\sqrt{n}).$$

Therefore, we can use Lemma 2.1 to obtain the inequality $t \cdot s^q \geq n^{\frac{\sqrt{nq}}{20}}$ (as $t \leq 2^{n^{1/3}} < e^{n/96}$ for large enough n). Plugging in $q = \lceil 15n^{1/3} \rceil$, we see that

$$s \geq \frac{n^{\frac{\sqrt{n/q}}{20}}}{t^{1/q}} \geq \frac{n^{\frac{n^{1/3}}{80}}}{2^{1/15}} = \exp(\Omega(n^{1/3})).$$

□

Proof of Theorem 0.3. We consider cases as follows:

Case $t = \text{poly}(n)$: Suppose there is a constant c such that $t \leq n^c$. Set $q = 20 \ln(2n^c\sqrt{d})$ and note that $q = \Theta(\log n)$. We see that by Lemma 2.1,

$$s \geq \frac{n^{\frac{\sqrt{d/q}}{20}}}{t^{1/q}} \geq n^{\frac{1}{20}\sqrt{\frac{d}{q}} - \frac{c}{q}}.$$

Note that $c/q < 1$ and decays to zero as d becomes larger. Furthermore, as $d = \omega(\log n)$ by assumption and $q = \Theta(\log n)$, $d/q = \omega(1)$. We conclude that if t is bounded by a polynomial in n, then s must be super-polynomial in n.

Case $s = \text{poly}(n)$: Suppose there is a fixed constant $c \geq 1$ such that $s \leq n^c$. Assume that $t < e^{d/96c^2}$, and set $q = d/1600c^2$. Then q indeed lies in the range to apply Lemma 2.1. We obtain the inequality

$$t \geq \left(\frac{n^{\frac{\sqrt{d/q}}{20}}}{s} \right)^q \geq n^{\frac{\sqrt{dq}}{20} - cq} = n^{cq} = n^{\Omega(d)},$$

which contradicts the assumption that $t < e^{d/96c^2}$. Hence, $t \geq e^{d/96c^2} = 2^{\Omega(d)}$ which is indeed super-polynomial in n whenever $d = \omega(\log n)$.

Thus, in either case, it is shown that both s and t cannot be polynomially bounded. Hence, the total-width of any $\sum$smABP computing $F_{n,d}$ cannot be polynomially bounded. □

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